



Produce your own insulin

A stem cell-based treatment given through an implant can create insulin in Type-1 diabetes patients. <https://digit.in/dec21-43>



Improving on power outages

Microgrids help communities to keep critical services running by generating their own electricity. <https://digit.in/dec21-44>

out for oxygen molecules from the injected molecules which leaves a mixture of hydrogen and carbon monoxide called syngas. This syngas is then funnelled into a second reactor where the syngas is converted into kerosene or methanol molecules.

The first trickle of fuel in the Móstoles refinery was announced in the month of June in 2019.

To test the concept, the researchers had set up a small 5kW pilot system on the roof of a building which used to run for 7 hours a day in intermittent sunlight. The device was able to produce 32 ml of methanol each day.

We know that's not a whole lot, but this itself demonstrates that the technology works and it could be scaled

says Steinfeld. "Can you imagine a Boeing or Airbus flying trans-Atlantic on batteries?"

ADVANTAGES OF USING SOLAR PRODUCED HYDROCARBON FUELS

- Can provide a carbon neutral source of fuels that can be a replacement to fossil fuels.
- Can be a much more sustainable source of energy
- Given it's preferable location, it can be set up on otherwise infertile soil.
- The presence of plants can give a source of work and income for the locals of otherwise poorly developed places.



Deserts are ideal locations to setup solar kerosene plants

up to commercial production and a large scale plant could look somewhat like a solar thermal power plant with fields of concentrators focusing sunlight on one single central tower.

According to calculations made by the team, a plant using 10 such fields, each collecting 100MW of solar radiative power, could produce 95,000 L of kerosene per day which is enough to get an Airbus A350 from London to New York and back again.

THE NEED FOR CONVERTING SOLAR ENERGY INTO FUEL

The electric battery technology we have as of now just doesn't pack the punch necessary to actually drive heavy industries such as sea freight and air travel.

"There is no way around jet fuel for long-haul commercial aviation,"

DISADVANTAGES OF USING SOLAR PRODUCED HYDROCARBON FUELS

- It would be way too expensive as an initial fuel source costing upto 9 USD or about 680 INR per gallon if sold to consumers.
- There is a high upfront cost in setting up the plants which would require monetary assistance in the form of subsidies and it would also require political support in order to get them off the ground which can limit its viability.

HOW IT CAN ALL COME INTO PLAY INTO THE REAL WORLD

Solar kerosene has the potential to slot into existing global infrastructure for storing, transporting and using fossil fuels. Therefore, the only problem that remains is its

high initial cost for consumers and the high upfront cost of setting up the solar plants required to produce solar kerosene.

However, the cost per gallon of the solar produced kerosene should fall as technology improves and more research is conducted to boost the efficiency of the solar refineries.

We could say the revolution is already underway as in May of 2019, a major airport in the Netherlands announced that it would host the world's first plant turning atmospheric carbon dioxide into liquid fuel.

In all of these projects, the modules that will capture the carbon dioxide from the air are being provided by Climeworks, which is a company spun out from Steinfeld's lab back in 2009.

Although we have a plan on how to get the technical side of it running, bringing solar kerosene into play won't be so easy without receiving political support from governments.

According to Johan Lilliestam, "The European Union's existing support instruments – emissions trading and offsetting – are not sufficient to stimulate market demand for solar fuels. In view of this, we propose the adoption of a European technology-specific quota system for aviation fuel. This would require airlines to acquire a specific share of their fuel from solar sources"

Whereas the study's authors recommend a share of 0.1 percent in the earliest phase of market adoption when the price of solar kerosene would be high and production capacities low. This would have little to no impact on the cost of flying but would still promote the construction of production facilities and set in motion a learning curve that could lead to the technological advances required for solar kerosene to be produced with better efficiency and can be sold at a lower price. The 0.1 percent quota could gradually be increased until solar kerosene achieves a market breakthrough without further support measures. 